



Volume-targeted ventilation

Martin Keszler *

Georgetown University Hospital, 3800 Reservoir Road, NW, Washington, DC 20007, United States

KEYWORDS

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Tidal volume;
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Abstract Recognition that volume, not pressure, is the key factor in ventilator-induced lung injury and the association of hypocarbia and brain injury dictate the need to better control delivered tidal volume. Volume-controlled ventilation, though much improved, still suffers from loss of volume due to endotracheal tube leak and gas compression in the circuit. Recent microprocessor-based modifications of pressure-limited, time-cycled ventilators combine advantages of pressure-limited ventilation with the ability to deliver a more consistent tidal volume. Each of the modes has advantages and disadvantages, with limited data available to judge their effectiveness. The Volume Guarantee mode, studied most thoroughly, provides automatic weaning of peak pressure in response to improving lung compliance and respiratory effort. More consistent tidal volume, fewer excessively large breaths, lower peak pressure, less hypocarbia and lower levels of inflammatory cytokines have been documented. It remains to be seen if these short-term benefits translate into shorter duration of ventilation or reduced incidence of chronic lung disease.

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1. Introduction

Advances in newborn respiratory support, along with widespread use of antenatal steroids and exogenous surfactant, have dramatically reduced mortality from acute respiratory failure. Early neonatal mortality is now more often from non-respiratory causes: infection, necrotizing enterocolitis and intracranial hemorrhage. The focus of respiratory support has thus shifted from reducing mortality to reducing the still unacceptably high incidence of chronic lung disease, due in part to improved survival of extremely preterm infants.

Based on early primate studies, high-frequency ventilation showed great promise in this regard, but the results of clinical trials have been inconsistent. Several recent trials failed to show any reduction in chronic lung disease, perhaps because of less injurious conventional ventilation in the control patients. Furthermore, on-going concerns about the hazards of inad-

vertent hyperventilation have limited acceptance of high-frequency ventilation as first-line therapy in infants with uncomplicated RDS. At the same time, technologically advanced synchronized ventilation modes have become widely available. The most exciting development in neonatal respiratory support is the advent of volume-targeted modalities of conventional ventilation that, for the first time, allow effective control of delivered tidal volume when ventilating even extremely low birth weight infants. In the following paragraphs, I will outline the rationale for volume-targeted ventilation, describe the different modes of volume-targeted ventilation, summarize available clinical studies and discuss the clinical application of Volume Guarantee, the most extensively studied mode of volume-targeted ventilation.

2. Rationale for volume-targeted ventilation

Because initial attempts to use traditional volume-controlled ventilation proved to be impractical in sick newborn infants, pressure-limited, time-cycled, continuous flow ventilation

* Tel.: +1 202 444 8569; fax: +1 202 784 4747.
E-mail address: keszlerm@gunet.georgetown.edu.

became the standard neonatal ventilatory mode more than 30 years ago. The ability to directly control inspiratory pressure has been considered for a long time to be a key advantage of pressure-limited over volume-controlled ventilation. This pre-occupation with inspiratory pressure as the chief culprit in ventilator-induced lung injury (VILI) and airleak remains a constant theme in neonatal respiratory support even today, despite mounting evidence that excessive volume, rather than pressure, is the critical determinant of VILI. Dreyfuss and colleagues demonstrated as early as 1988 that severe acute lung injury occurred in animals ventilated with large tidal volume, regardless of whether that volume was generated by positive or negative inspiratory pressure. On the other hand, animals whose chest wall and diaphragmatic excursion were limited by external binding, but who were exposed to the same high inspiratory pressure experienced much less acute lung damage [1]. This work and other similar experiments clearly show that excessive tidal volume, not pressure per se, is chiefly responsible for lung injury [2,3]. However, it has been only recently that full recognition of the importance of volutrauma has rekindled interest in directly controlling tidal volume.

Perhaps an even more compelling reason for volume-targeted ventilation is the extensive body of evidence documenting that hypocapnia is associated with neonatal brain injury [4,5]. Despite increasing awareness of the dangers of hypocapnia, inadvertent hyperventilation remains a common problem with pressure-limited ventilation, especially early in the clinical course when lung compliance changes rapidly in response to clearing of lung fluid, surfactant administration and optimization of lung volume. A recent study from England demonstrated that 30% of ventilated infants had at least one blood gas with $\text{PaCO}_2 < 3.3$ kPa during the first day of life [6].

2.1. Limitations of traditional volume-controlled ventilation

Traditional volume-controlled ventilation was attempted in newborn infants in the early 1980s using the relatively crude devices then available, but proved to be impractical and was soon abandoned. More recent refinements in the technology, ability to measure V_T at the airway opening and better understanding of the mechanics of gas delivery have made it possible to utilize volume-controlled ventilation in premature newborn infants. In volume-controlled ventilation, the device delivers a set volume of gas into the ventilator circuit. Pressure rises passively, inversely related to lung compliance, as gas enters the lungs. A portion of the volume delivered into the circuit is lost to compression of gas in the circuit and humidifier and to stretching of the elastic tubing. When the circuit volume and compliance are low in relation to the patient's lungs, this is a relatively minor problem and can be readily compensated using standard nomograms. However, in small neonates with poorly compliant lungs, the volume loss to the circuit/humidifier is proportionally much larger and more difficult to compensate. More importantly, there is often a large and unpredictable loss of tidal volume to the ubiquitous and highly variable leak around uncuffed endotracheal tubes, which cannot be easily compensated. It is important to be aware of these limitations, because a number of the newer ventilators that claim to be suitable for both neonates and adults have such traditional volume-

controlled modes available, although in some devices the lowest possible V_T setting is far greater than what is appropriate in small preterm infants. Nonetheless, a recent study demonstrated that when significant endotracheal leak is eliminated and the device is used with a proximal flow sensor that allows monitoring of exhaled tidal volume at the airway opening and frequent adjustment of the set tidal volume (usually requiring 8–10 ml/kg set V_T to deliver 4–6 ml/kg at the airway opening) these modes can be effectively employed in small newborn infants [7].

3. Neonatal volume-targeted ventilation

A number of modifications of time-cycled, pressure-limited ventilation have been developed in recent years, to combine the advantages of pressure-limited ventilation with the benefits of controlling delivered tidal volume. These modes are designed to deliver a target tidal volume by micro-processor-directed adjustments of inspiratory pressure or time. Each of the available modes has advantages and disadvantages. The modes of volume-targeted ventilation used most commonly will be discussed below.

3.1. Volume limit

Volume limit is a function of the Bear Cub 750 PSV (Viasis Medical Systems Conshohocken, PA). This is not true volume-targeted ventilation, as the only enhancement over simple pressure-limited ventilation is a volume limit setting; when the limit V_T is exceeded, the device terminates inspiration thus avoiding excessive tidal volume. There is no automatic adjustment of inspiratory pressure and no provision to ensure that adequate V_T is delivered when compliance or patient effort decreases. The reliance on inspiratory V_T measurement means that significant leak around the ETT may lead to inadequate V_T delivery.

3.2. Targeted tidal volume (TTV)

This is in essence a simple volume limit function available on the SLE 5000 (Specialised Laboratory Equipment Ltd, South Croydon, UK). The device increases the rise time of the pressure waveform to improve the chance of effectively limiting V_T to the desired target. When the volume limit function is turned off, as a safety measure, the PIP automatically drops to 5 mbar above the PEEP to avoid potentially excessive V_T due to inappropriately high PIP; the user must then actively adjust the PIP. The same limitations of Volume Limit apply: there is no automatic adjustment of inspiratory pressure and no provision to ensure that adequate V_T is delivered when compliance or patient effort decreases. Reliance on inspiratory V_T measurement may lead to inadequate V_T delivery with significant leak around the ETT. No clinical data evaluating this feature have been published to date.

3.3. Pressure regulated volume control (PRVC)

PRVC is a pressure-limited time-cycled assist/control mode that adjusts inspiratory pressure to target a set tidal volume, based on the tidal volume of the previous breath. Breath to breath increment is limited to 3 cm H_2O , up to 5 cm H_2O below the set upper pressure limit. Because pressure adjustment is

based on the previous breath, variable patient respiratory effort will cause fluctuations in delivered V_T . The main problem with the PRVC mode of the Maquet Servo 300 and the newer Servo-i (Maquet Inc., Bridgewater, NJ, formerly Siemens, Solna, Sweden) for newborn infants is the major inaccuracy of V_T measurement performed at the ventilator end of the circuit, rather than at the airway opening [8]. Recently, a second flow sensor that permits monitoring of actual V_T has been made available, though the V_T regulation is still based on the volume measured at the ventilator outlet.

3.4. Volume assured pressure support (VAPS)

The Volume Assured Pressure Support mode on the Bird VIP Gold (Viasis Medical Systems Conshohocken, PA) is a hybrid mode, which seeks to ensure that the targeted V_T is reached. Each breath starts as a pressure-limited breath, but if the set V_T is not reached, the breath converts to a flow-cycled mode by prolonging the inspiratory time with a passive increase in peak pressure. This may result in a rather prolonged inspiratory time leading to expiratory asynchrony. Targeting tidal volume based on inspiratory tidal volume is susceptible to underventilation and the presence of significant endotracheal tube leak. Furthermore, there is no provision for automatically lowering inspiratory pressure as lung compliance improves. Therefore, the focus is on ensuring a large enough V_T , but no provision is made to avoid excessive tidal volume and inadvertent hyperventilation, or to allow for automatic weaning.

The new Avea ventilator by Viasis shares the basic features of VAPS, but adds a Volume Limit function that will terminate inspiration if the upper limit of tidal volume is exceeded. The algorithm used to assure delivery of the target V_T has been refined to try to avoid excessive inspiratory time. The ventilator calculates the decelerating inspiratory flow required to deliver the set volume in the set inspiratory time. When a pressure control breath is delivered and peak flow decelerates to this calculated peak inspiratory flow, if the set volume has not been delivered, the ventilator will automatically transition to a continuous flow mode until the set volume has been delivered (volume cycling). If the set volume is met or exceeded during delivery of the pressure control breath, the ventilator will complete the breath as a normal Pressure Control breath.

The addition of a volume limit should reduce the risk of volutrauma and hyperventilation, but still does not provide for automatic weaning of inspiratory pressure. No clinical studies are available at this time to validate the performance of this device in newborn infants. As with the other approaches, the interaction of an actively breathing infant with the device will make the achievement of a stable V_T much more difficult than in bench tests.

3.5. Volume Guarantee (VG)

The Draeger Babylog 8000-plus (Dräger, Lübeck, Germany) offers a Volume Guarantee (VG) option that may be combined with any of the standard ventilator modes (AC, SIMV, PSV). The VG mode is a volume-targeted, time- or flow-cycled, pressure-limited form of ventilation. The operator chooses a target tidal volume and selects a pressure limit up to which the ventilator operating pressure (the working pressure) may be adjusted

(Fig. 1). The microprocessor compares the tidal volume of the previous breath, using exhaled tidal volume to minimize possible artifact due to airleak, and adjusts the working pressure up or down to try to achieve the set tidal volume. The algorithm limits the pressure increment from one breath to the next to 3 cm H₂O, in order to avoid over-correction leading to excessive tidal volume. This, and the fact that the exhaled tidal volume of the prior breath is used, means that with very rapid changes in compliance or patient inspiratory effort, several breaths may be needed to reach target tidal volume. In order to minimize the risk of excessively large tidal volume, the microprocessor opens the expiratory valve, terminating any additional gas delivery if the inspired tidal volume exceeds 130% of the previous breath. The algorithm is designed to make slower incremental adjustment for low tidal volume and more rapid adjustment for excessive, potentially dangerous tidal volume. The auto-regulation of inspiratory pressure makes Volume Guarantee a self-weaning mode. Because weaning occurs in real-time, rather than intermittently in response to blood gases or intermittent observation of delivered tidal volume, the VG mode has the potential to achieve faster weaning from mechanical ventilation. Though more tolerant of endotracheal tube leak than other volume-targeted modes because of the use of exhaled tidal volume measurement, VG becomes impractical in the presence of leak that substantially exceeds 40%, because the V_T measurement increasingly underestimates the true value. The problem is easily corrected by re-intubating the infant with a larger endotracheal tube.

4. Clinical studies of volume-targeted ventilation

4.1. Volume control ventilation (VC)

Two studies have been conducted with VC by investigators from Middlesbrough, England, both using the Bird VIP ventilator and

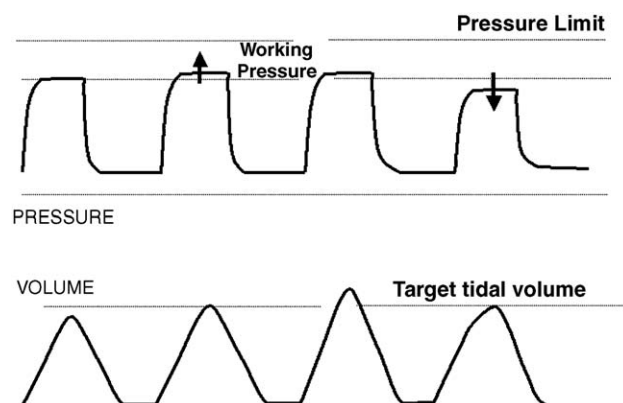


Figure 1 Principles of operation of Volume Guarantee. The device auto-regulates the PIP (working pressure) within preset limit (pressure limit) to achieve tidal volume that is set by the user. Regulation of PIP is in response to *exhaled* tidal volume to eliminate artifact due to endotracheal tube leak. As illustrated in the figure, the first breath falls short of the target V_T , leading to an increase in working pressure of the next breath. The second breath is on target, thus there is no change in working pressure. The third breath is above the target, which results in a drop in working pressure for the fourth breath, bringing the V_T back to the target value.

both using time to achieve $\text{AaDO}_2 < 13$ kPa (100 Torr) or mean airway pressure < 8 cm H_2O , maintained for > 12 h as the primary endpoint. The first study, published in 1997, included 50 infants with birth weight > 1200 g (actual mean birth weight was nearly 1800 g) [9]. Tidal volume of 5–8 ml/kg was targeted in both groups. Infants randomized to VC reached success criteria faster and had shorter duration of mechanical ventilation (122 ± 65 h for VC vs 162 ± 134 h). There was a trend toward less BPD and IVH in the VC group. These encouraging results demonstrated feasibility of this mode of ventilation in larger preterm infants, but left the question unanswered as to whether this modality is useful in the much smaller infants who today constitute the large majority of ventilated infants.

The newer model of the VIP Bird Gold that allows lower V_T settings and accurate monitoring of exhaled V_T at the airway opening made it possible to address this question. In a study just published in the *Journal of Pediatrics*, Singh et al. randomly assigned 109 preterm infants 24–31 weeks and 600–1500 g to VC or to time-cycled, pressure-limited ventilation [7]. Exhaled V_T of 4–6 ml was targeted in both groups. The investigators found no difference in the primary outcome of reaching $\text{AaDO}_2 < 13$ kPa (100 Torr) or mean airway pressure < 8 cm H_2O in the overall group, but a post-hoc analysis showed faster weaning from VC in infants < 1000 g and those with more severe respiratory failure. There was no difference in the duration of mechanical ventilation, oxygen requirement or other secondary outcomes. Though the results did not clearly show superiority of VC, the demonstrated feasibility of VC even in extremely low birth weight infants is very promising and should encourage further evaluation of this strategy.

4.2. PRVC

D'Angio and colleagues studied 212 preterm infants (birth weight of 500 to 1249 g) randomly assigned to PRVC or to pressure-limited synchronized intermittent mandatory ventilation (SIMV) [10]. Infants were to receive their assigned mode of ventilation until extubation or death. Crossover was allowed upon meeting predetermined failure criteria. Mean birth weights were similar in the SIMV (888 ± 199 g) and PRVC (884 ± 203 g) groups. No differences were detected between SIMV and PRVC groups in the primary outcome, the proportion of infants alive and extubated at 14 days (41% vs 37%, respectively), length of mechanical ventilation in survivors (median, 24 days vs 33 days, respectively), or the proportion of infants alive without supplemental oxygen at 36 weeks' postmenstrual age (57% vs 63%, respectively). More infants crossed over from PRVC than from SIMV. The authors concluded that PRVC offered no demonstrable advantage over SIMV.

4.3. Volume assured pressure support

No published clinical trials are available.

4.4. Volume Guarantee

Cheema et al. examined the feasibility and efficacy of Volume Guarantee in 40 premature newborn infants [11]. In a four-hour crossover trial they compared AC with and without VG in infants with acute RDS, and SIMV with and without VG during the weaning phase. In both VG groups, infants were

able to achieve equivalent gas exchange using lower peak airway pressure. There were fewer excessively large tidal volumes during the Volume Guarantee periods. The authors concluded the VG mode was feasible and may offer the benefit of lower airway pressures. Due to the short duration of the study, no major conclusions could be drawn, other than that the ventilator performs as intended and no short-term adverse effects were evident.

Herrera et al. compared the effects of SIMV+VG with SIMV alone on ventilation and gas exchange in a group of very low birth weight infants recovering from acute respiratory failure [12]. They showed that short-term use of SIMV+VG resulted in automatic reduction of the mechanical support and enhancement of the spontaneous respiratory effort, while maintaining gas exchange relatively unchanged in comparison to SIMV alone. Further shift of the work of breathing to the patient was observed when the target V_T was reduced from the normal 4.5 ml/kg target to 3 ml/kg. The proportion of excessively large breaths > 7 ml/kg was reduced from 16% with SIMV to 6% with SIMV+VG. The study added further confirmation of the short-term benefit and potential of Volume Guarantee.

In a small prospective clinical trial involving 34 fairly large preterm infants (mean birth weight 1122 g), Nafday compared SIMV to PSV+VG over the first 24 h of life [13]. They did not demonstrate any difference in time to extubation or other important clinical outcomes, but this "pilot" study lacked adequate statistical power. The authors noted that mean airway pressure declined more rapidly in the SIMV group, but this is simply the result of lowering the ventilator rate (leading to lower I:E ratio) with SIMV. There was no difference in the rate of decline of peak pressure. Significantly fewer blood gases were needed in the Volume Guarantee group.

Our group earlier showed that Volume Guarantee combined with AC, SIMV or PSV in a short-term crossover study led to significantly less variability of tidal volume with VG, when compared to AC, PSV or SIMV alone and that peak inspiratory pressures were similar [14]. In a subsequent randomized clinical trial, we demonstrated that, when combined with the AC mode, Volume Guarantee maintained PaCO_2 and V_T within a target range more consistently than assist/control alone during the first 72 h of life in preterm infants with uncomplicated respiratory distress syndrome (Fig. 2) [15]. The first paper documented that the device functions as intended in the clinical setting with the anticipated reduction of the variability of tidal volume, even in actively breathing infants. The prospective trial demonstrated that excessively large tidal volume and hypocarbia could be reduced, though not eliminated, with the use of Volume Guarantee. These findings suggest the potential of VG to reduce many of the important adverse effects of mechanical ventilation.

Next, in a short-term crossover trial involving extremely low birth weight infants (679 ± 138 g) we addressed the question of whether VG is more effective when combined with AC or SIMV [16]. As anticipated, the tidal volume was more stable when Volume Guarantee was combined with AC, because the interval between breaths is longer during SIMV, leading to slower adjustment. An unanticipated finding was that during SIMV, the infants had lower and more variable oxygen saturation, and were significantly more tachycardic and tachypneic. By design, the tidal volume was identical,

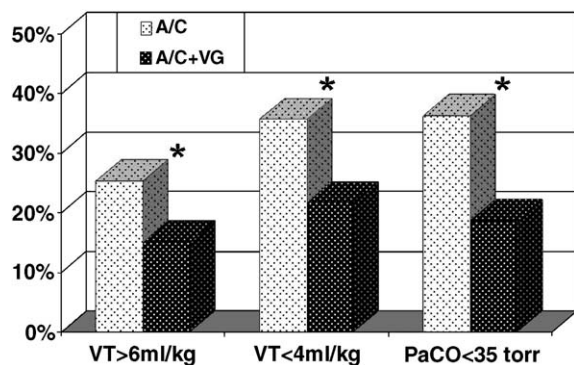


Figure 2 Proportion of breaths with tidal volume above (left panel) and below (middle panel) target range of 4 to 6 ml/kg. The right panel shows proportion of blood gases with $\text{PaCO}_2 < 35$ Torr. $*p < 0.001$.

but significantly higher PIP was required during SIMV to achieve the same tidal volume. The tachypnea, tachycardia and lower oxygen saturation suggest that the reason for the higher machine PIP was that the infants were tiring during the SIMV period and contributing less to their ventilation by the end of the 2 hour period when the measurements were obtained (recall that during synchronized ventilation, the delivered tidal volume is the result of the combined effort of the baby and the ventilator pressure; if the baby contributes less, the ventilator needs to generate higher PIP to deliver the same V_T).

To address the concern regarding possible untoward effects of the additional instrumental dead-space (IDS) of the flow sensor and to establish normative data for target V_T in extremely low birth weight infants, we reviewed 451 paired observations of V_T and arterial blood gas measurements in 47 infants < 800 g at birth (mean wt 627 g, range 400–790 g, data not yet published). The V_T/kg needed for normocapnia was inversely related to weight ($r = -0.56$, $p < 0.01$), indicating some effect of the fixed IDS (Fig. 3). Mean V_T/kg of infants ≤ 500 g was 5.7 ± 0.36 ml compared to 4.7 ± 0.39 ml for those ≥ 700 g ($p < 0.001$). However, the absolute mean set and measured V_T were 3.12 ± 0.76 ml and 3.13 ± 0.66 ml, respectively, barely above the estimated instrumental + anatomical dead-space of 3 ml. While maintaining normocapnia, 48% of all tidal volumes were below the estimated dead-space. We concluded that there is a greater impact of IDS in the tiniest infants but there is no need to forgo synchronized and VG ventilation due to concerns about IDS. Effective alveolar ventilation occurs with V_T at or below dead-space, suggesting that a spike of fresh gas penetrates through the dead-space gas.

Finally, in what is probably the most convincing evidence to date about the potential benefits of volume-targeted ventilation, Lista et al. randomly assigned 53 preterm infants with RDS to Volume Guarantee combined with PSV or to PSV alone, using target tidal volume of 5 ml/kg [17]. He demonstrated decreased levels of pro-inflammatory cytokines and a trend toward shorter duration of mechanical ventilation in the VG group. These data clearly support the hypothesis that volume-targeted ventilation is capable of reducing ventilator-induced lung injury, which is mediated by the release of pro-inflammatory cytokines in a process known as biotrauma [3].

Interestingly, a subsequent similar study by the same authors now using a target tidal volume of 3 ml/kg showed an increase in pro-inflammatory cytokines [18], most likely as a consequence of atelectasis that resulted from the combination of low tidal volume and low end-expiratory pressure of 3–4 cm H_2O that was used [19].

4.5. Importance of open lung strategy

As a result of the overwhelming evidence that excessive tidal volume, rather than high inspiratory pressure, is the primary determinant of lung injury, most clinicians now at least monitor the delivered tidal volume when using pressure-limited ventilation or use one of the forms of volume-targeted ventilation. The critical importance of distributing this tidal volume evenly into an optimally aerated lung has not been as widely appreciated and requires special emphasis. As can be seen in Fig. 4, if extensive atelectasis is allowed to persist, the normal, physiologic tidal volume entering the small proportion of open alveoli will inevitably lead to over-expansion of this relatively healthy portion of the lung with subsequent volutrauma/biotrauma. The collapsed portion of the lung is also being damaged as a result of a process for which the term 'atelectotrauma' has been coined. Exudation of protein rich fluid with increased surfactant inactivation and release of inflammatory mediators has been documented in the atelectatic areas. Shear forces and uneven stress in areas where atelectasis and overinflation coexist add to the damage. *Thus, the benefits of volume-targeted ventilation cannot be realized without ensuring that this tidal volume is distributed evenly throughout the lungs!*

In practical terms, this "open lung concept" is achieved by applying adequate positive end-expiratory pressure (PEEP). For a variety of reasons, including poorly conceived animal studies where moderate to high levels of PEEP were applied to animals with normal (i.e. very compliant) lungs, resulting in significant hemodynamic impairment, there is widespread fear of using adequate levels of end-expiratory pressure. This "PEEP-o-phobia" is only slowly being overcome and remains one of the most important obstacles to optimizing the way mechanical ventilation is practiced. Finally, it is important to understand that there is no single "safe" PEEP level. Rather, optimal end-expiratory pressure must be tailored to the degree of lung injury (i.e. lung compliance). For infants with

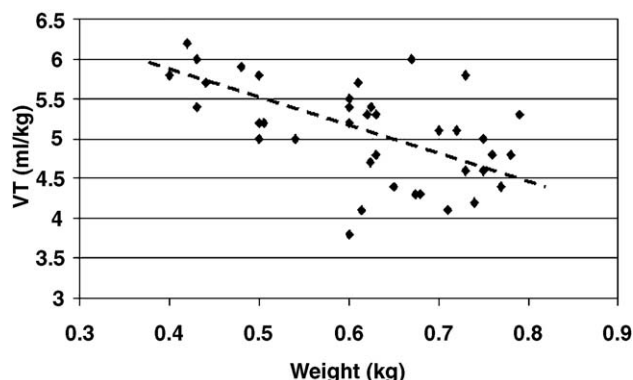


Figure 3 Relationship between birth weight and V_T/kg required for normocapnia in extremely low birth weight infants ($r = -0.56$, $p < 0.01$).

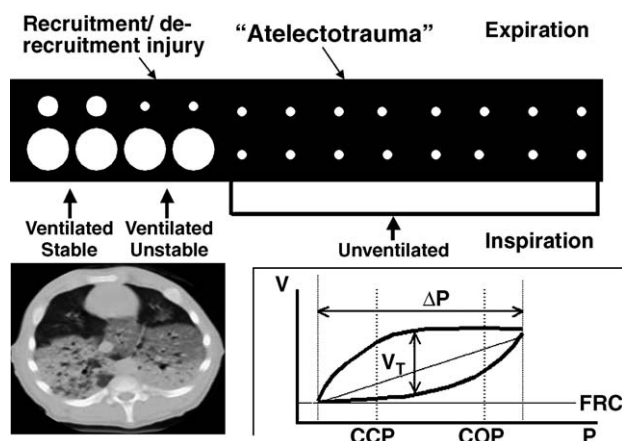


Figure 4 Non-homogeneous aeration in RDS. Extensive atelectasis is often present in surfactant deficient lungs as illustrated on the Computer Tomography view in lower left corner (courtesy of Anastasia Pellicano, MD). This situation is schematically represented in the middle panel. The corresponding pressure–volume loop is shown in the lower right portion of the figure. In the absence of extrapulmonary right-to-left shunting, poor oxygenation strongly suggests that atelectasis is present. If this situation is allowed to persist, a normal, physiologic tidal volume entering only the small proportion of open alveoli will inevitably lead to over-expansion and lung damage.

healthy lungs and thus normal lung compliance, PEEP of 3 cm H₂O is adequate and PEEP of 5 cm H₂O may result in over-expansion of the lungs with impairment of venous return, elevated cerebral and systemic venous pressures and decreased cardiac output. On the other hand, severely atelectatic, poorly compliant lungs may require PEEP levels of 8–10 cm H₂O or more to achieve adequate alveolar recruitment and improve ventilation/perfusion ratio. Because we seldom ventilate infants with healthy lungs, PEEP of <5 cm H₂O should be the exception, rather than the rule.

5. Conclusion

Several forms of volume-targeted ventilation have been shown to be feasible and safe even in extremely low birth weight infants, the population that now dominates our NICUs and is at greatest risk of developing chronic lung disease. Volume Guarantee ventilation has been demonstrated to function as intended and to lead to more stable tidal volume with lower incidence of hypocarbia and excessively large tidal volumes. When combined with other lung-protective strategies aimed at optimizing lung volume and ensuring even distribution of tidal volume, volume-targeted ventilation appears to offer the best hope of making a significant impact on ventilator-induced lung injury. However, definitive evidence of major clinical benefit of volume-targeted ventilation is not available at this time. It must be emphasized that the development of chronic lung disease in extremely preterm infants is multifactorial. The degree of prematurity and consequences of intrauterine inflammation have a very large effect, likely to dwarf any impact of ventilation strategy. Consequently, large multicenter studies involving 500 to 800 infants will be needed to have sufficient statistical power to detect modest, but clinically important differences in the incidence of chronic lung disease. A surrogate

outcome, such as more rapid weaning from mechanical ventilation, is likely to be an easier endpoint to demonstrate, because of the self-weaning nature of Volume Guarantee.

Appendix A. Volume Guarantee: clinical guidelines

N.B.: Though volume-targeted ventilation modes share certain characteristics, each device functions differently. Although many of the principles discussed below apply to volume-targeted ventilation in general, these guidelines are specific to the Volume Guarantee mode of the Draeger Babylog. Please, consult product literature of each manufacturer for specific clinical guidelines for their respective ventilators.

A.1. Initiation

- Optimally, Volume Guarantee should be initiated as soon as possible after intubation, because this is the time when most rapid changes in lung compliance occur.
- Assist control or pressure support is the preferred basic modes, there is no need to switch to SIMV for weaning.
- The usual starting target V_T is 4.5 ml/kg during the acute phase of RDS in most infants. Babies with birth weight less than approximately 750 g will require 5 to 6 ml/kg, because the modest additional dead-space of the flow sensor becomes proportionally more important in the smallest of infants.
- Larger V_T is also needed in older infants with chronic lung disease because of increased anatomical dead-space due to stretching of the trachea (acquired tracheomegaly) and increased physiologic (also known as alveolar) dead-space [wasted ventilation due to poor ventilation/perfusion matching]. Tidal volume of 6–8 ml/kg is often needed to achieve PaCO₂ in the high 50 s in infants with moderate to severe chronic lung disease.
- Infants with meconium aspiration also may require larger V_T because of increased alveolar dead-space.
- Infants with pulmonary hypoplasia (such as congenital diaphragmatic hernia) require correspondingly smaller V_T , typically 3 to 3.5 ml/kg.
- The PIP should be set initially at a pressure about 5 cm H₂O above that estimated to be sufficient to deliver a normal tidal volume. If the target V_T cannot be reached with this setting, increase the pressure limit until the desired V_T is generated. *N.B: Make sure the endotracheal tube is not in the mainstem bronchus or obstructed on the carina!*
- Once the working pressure stabilizes, set the pressure limit about 20% above the average working pressure displayed on the front panel, in order to give the device adequate room to adjust the working pressure.
- It is important to record not only the PIP limit but also the working pressure, so that the level of support the infant is actually receiving is clearly evident.

A.2. Subsequent adjustments

- Subsequent adjustment to target tidal volume may be needed, based on PaCO₂, though this is seldom necessary. The usual increment is 0.5 ml/kg.

- The PIP limit needs to be adjusted from time to time (usual increment 2–3 cm H₂O) to keep the PIP limit sufficiently close to the working pressure and at the same time avoid frequent alarms.
- *Please, note:* If the flow sensor is temporarily removed (such as around the time of surfactant administration or delivery of nebulized medication), if its function is affected by reflux of secretions or surfactant, or it malfunctions for any reason, the working pressure will default to the PIP limit. The manual inspiration (effected by depressing the key on the front panel) also uses the set pressure limit. Therefore, it is important to keep the PIP limit sufficiently close to the working pressure to avoid volutrauma. Ideally, when removing the flow sensor for significant periods, such as when nebulizing medications, the PIP limit should be adjusted to match the average or recent working pressures.
- If the infant is persistently tachypneic (RR > 80), consider increasing the tidal volume target even if the PaCO₂ and pH are normal, because this suggests the infant's work of breathing is excessive. In this situation, the infant is often generating V_T greater than the set target and getting very little help from the ventilator. (However, if the PaCO₂ is low and the RR is high, sedation may be indicated.)
- If the pressure limit has to be increased substantially and/or repeatedly, verify that the tidal volume measurement is accurate (assess chest rise, obtain a blood gas) and, if true, seek the cause of the change in lung mechanics (examine the patient, obtain chest radiograph). *Please, note* that if the endotracheal tube enters the right mainstem bronchus, the ventilator will continue to attempt to deliver the entire tidal volume, now into the single lung! It is critical that mainstem bronchus intubation is recognized promptly with any form of volume-targeted ventilation.

A.3. Weaning from mechanical ventilation

- When the target tidal volume is set at the low end of the normal range (usually 4 ml/kg) and PaCO₂ is allowed to rise the low to mid-40s, weaning occurs automatically ("self-weaning"), as the infant should have a good respiratory drive and gradually takes over the work of breathing.
- For infants who have chronic lung disease, a higher tidal volume should be used even during weaning. Self-weaning occurs as long as the pH is low enough to give the infant good respiratory drive.
- If the tidal volume is set too high and/or the PaCO₂ is too low, the baby will not have a respiratory drive and will not "self-wean". Instead, she will become dependent on the ventilator due to lack of respiratory muscle training.
- For obvious reasons, oversedation must be avoided during the weaning phase.
- If an infant appears to not be weaning as expected, despite apparently improving lung disease, try lowering tidal volume by 0.5 ml, as long as blood gases are adequate and the work of breathing does not appear excessive.
- If significant oxygen requirement persists, PEEP may need to be increased to keep mean airway pressure from falling as PIP is automatically lowered. Failure to do so may

result in atelectasis, worsening lung compliance and failure of extubation.

- Most infants can be extubated when they consistently maintain tidal volume at or above the target value with delivered PIP < 10–12 cm H₂O (< 12–15 cm H₂O in infants > 1 kg) with FiO₂ < 0.35 and good sustained respiratory effort.
- Observing the graphic display of the delivered PIP is helpful in assessing for periodic breathing (variable respiratory effort) and apnea that may require methylxanthine administration to facilitate extubation. Lowering the backup rate of A/C or PSV to 20 breaths/min may help unmask inconsistent respiratory effort.

A.4. Alarms/troubleshooting

The Volume Guarantee option will generate additional alarms, which may prove annoying when they are excessive. These alarms are generated because the device, in Volume Guarantee mode, provides continuous feedback regarding adequacy of ventilator support. Significant fall in lung compliance, decreased spontaneous respiratory effort, impending accidental extubation (manifested as increased endotracheal tube leak) and forced exhalation episodes all will all generate "low tidal volume" alarms. When properly used, this information improves care in these vulnerable infants. It is critical to evaluate the cause of repeated alarms and correct any correctable problems. Large leak results in underestimation of delivered V_T and triggers the low V_T alarm when the device is unable to reach the target V_T at the set PIP limit. When the leak exceeds about 40% the VG mode no longer functions reliably due to inability to accurately measure V_T; the problem is readily corrected by replacing the endotracheal tube by one of more appropriate size. Unnecessary alarms can be avoided by optimizing settings and alarm limits.

Use of longer alarm delay settings, appropriate pressure-limit settings, avoidance of large leak around endotracheal tubes and adequate physical comfort measures or sedation will minimize nuisance alarms.

If the low tidal volume alarm sounds repeatedly in the absence of excessive leak, increase the pressure limit, so as to allow the V_T to be reached *AND INVESTIGATE THE CAUSE* of the change in lung mechanics (e.g. right mainstem bronchus intubation, atelectasis, pneumothorax, pulmonary edema, etc.).

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